

Verifiable Human Oversight as European Digital Infrastructure

An open standard for Article 14 enforceability — owned by no single vendor, verifiable by anyone, offline.

The AI Act requires **human oversight** of high-risk AI (Article 14). But "a human was in the loop" is, today, an **unfalsifiable claim** — there is no artifact a regulator, court, or citizen can independently check that a specific, accountable person authorized a specific high-risk action before it executed. Europe is on course to require oversight it cannot verify — and to depend on US hyperscalers for the verification layer itself.

THE SOVEREIGNTY GAP

Whoever controls the layer that *proves* human oversight controls the audit substrate of European AI. If that layer is a proprietary feature inside a non-European cloud, oversight becomes something Europe *rents* rather than something it can *verify* for itself. The enforceability of Article 14 should not depend on a single vendor's API.

WHAT EMILIA PROTOCOL IS

EMILIA Protocol (EP) is an **open standard** (Apache-2.0, on the IETF standards track) for **authorization receipts**: before a high-risk action runs, a named human approves the *exact* action on their own device, and the system emits a signed evidence packet that **anyone can verify offline** — **with no account and without trusting the issuer's servers**. Alter one byte and verification fails. The receipt stays valid even if the issuing vendor disappears.

WHY THIS IS SOVEREIGN INFRASTRUCTURE, NOT A PRODUCT

- **No lock-in.** Open format, open verifier — any European institution can self-host, fork, or independently verify. The standard is permissionless.
- **Independently interoperable.** Three independent implementations (JavaScript, Python, Go) agree on a public conformance suite — the IETF bar for a real standard, not a single vendor's binary.
- **Offline and durable.** A receipt verified in 2026 still verifies in 2035 with only the receipt and the approver's public key — audit evidence that outlives any vendor.

EUROPE CAN ADOPT THE OPEN LAYER, NOT A BLACK BOX

A dozen efforts are now racing to define "a receipt for an AI agent's action." EP has specified and implemented the **thin layer that composes them** into a single, offline, verifiable decision — and supplies the one piece none of the others do: **a named human's authorization**. Europe's choice is between adopting that open, neutral composition layer or inheriting a proprietary one. The open layer is the sovereign choice.

A CONCRETE ARTICLE 14 CASE: HIGH-RISK MEDICAL AI

Medical AI is classified high-risk and must carry human oversight. EP expresses the clinician's authorization as **tamper-evident, offline-verifiable evidence** — mapping directly onto the independent double-check European hospitals already practice — and the receipt is **PHI-free by construction** (only hashed identifiers), so the evidence can be audited without a data-protection disclosure. The same primitive covers public procurement and benefits administration.

Open standard (Apache-2.0) · IETF Internet-Drafts (authorization receipts; multi-party quorum) · 3 interoperable verifiers (JS/Python/Go) over 8 cross-language conformance suites · independently re-verified by an outside implementer (2026) · machine-checked TLA+ & Alloy models · NIST AI RMF and EU AI Act Art. 9–15 + 26 mappings · try it: **`npx @emilia-protocol/crash-test`**

Iman Schrock — Founder, EMILIA Protocol · team@emiliaprotocol.ai · emiliaprotocol.ai · github.com/emiliaprotocol